Week 2 Session A

Tuesday, Jan 20 2026

Overview

- Review: AI, ML, (NN,) DL
  - Activity in Miro Board
- Generative AI
- Large Language Models
  - Famous LLMs
  - What is a model?
  - What is a language model?
  - What is a LLM?
  - Feeling how it works

Review: AI, ML, (NN,) DL

- Invite you to the Miro Board
- Once you all gained access, move items on the left to the right and reason your decisions

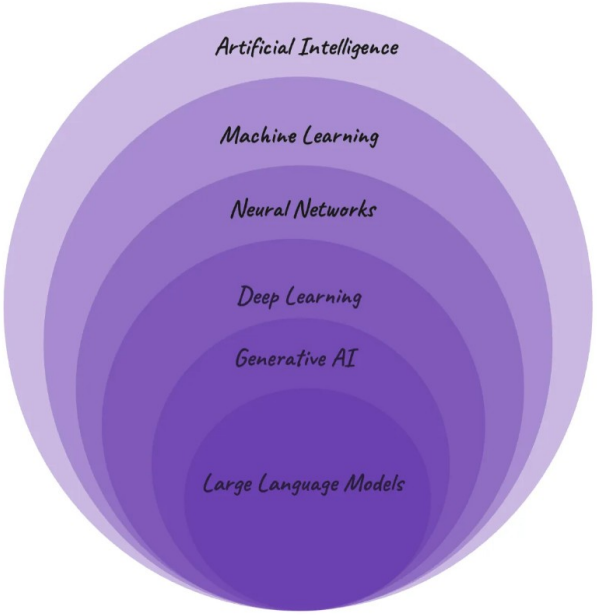
miro

Review: AI, ML, (NN,) DL

- Judge the following statements:
  - Machine learning is a subset of deep learning.
  - All machine learning is a form of artificial intelligence.
  - All artificial intelligence touch on deep learning.
  - Deep learning is a subset of machine learning.
  - All machine learning uses neural networks.

Generative AI

- Where does the concept “generative AI” fit into our visual?

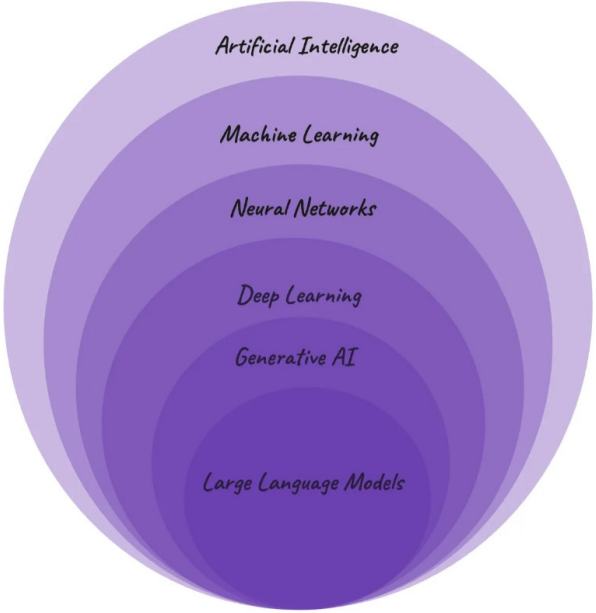


Generative AI are deep learning models that can **generate** high-quality text, images, and other content based on the data they were trained on.

- It generates.
- It’s creative in a **probabilistic** way.
- It comes from deep learning.

Large Language Models

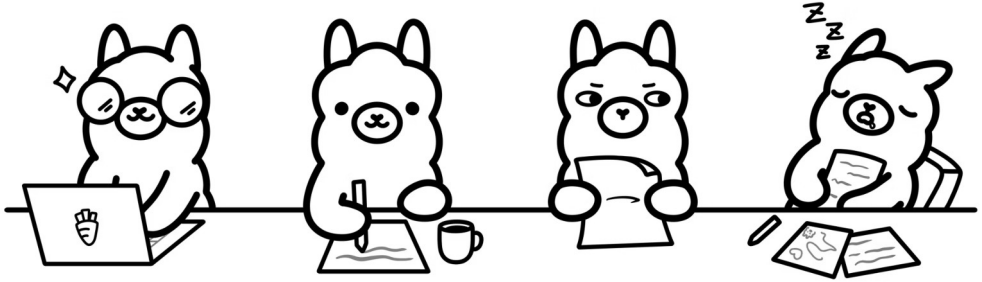
- Where does the concept “Large Language Models” fit into our visual?



- Judge the following statements:
  - Large Language Models are one specific type of Generative AI.
  - All LLMs are Generative AI, but not all Generative AI are LLMs.

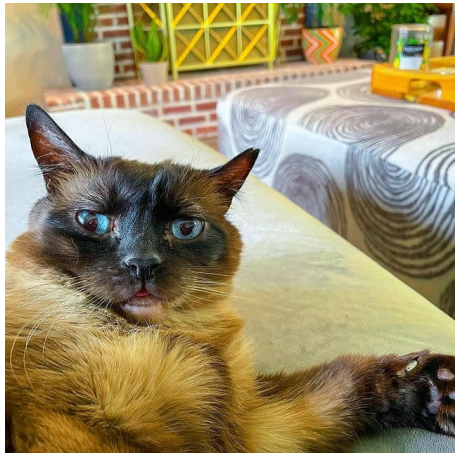
Large Language Models

- What are some of the popular LLMs?
  - Which companies own these models?
  - Where are they based?



LLMs Basics I

- What is a model? What is modeling?



LLMs Basics I

- What does a model look like?

```
Layer 1:  0.12, -0.04,  0.87,  1.23, -0.55, 0.03,  ...
Layer 2: -0.44,  0.15,  0.00,  0.98,  0.21, -0.11, ...
Layer 3:  0.95, -0.12,  0.22, -0.08,  0.67, 0.41,  ...
... (thousands or billions more parameters in large models)
```

Architecture

Parameters

LLMs Basics II

- What is a language model?

Guess the next word:

- This morning I spilled coffee on my…
- It’s so cold outside, remember to wear your …
- Titi is sick this week, so we are not meeting …
- Teddy has a pet and it is a …
- Rebecca comes from New York and people in Michigan always ask her why she decided to come …
- Theo studies journalism and we assume he enjoys reading and …

LLMs Basics II

- What does a language model look like?

```
Layer 1:  0.12, -0.04,  0.87,  1.23, -0.55, 0.03,  ...
Layer 2: -0.44,  0.15,  0.00,  0.98,  0.21, -0.11, ...
Layer 3:  0.95, -0.12,  0.22, -0.08,  0.67, 0.41,  ...
... (thousands or billions more parameters in large models)
```

LLMs Basics III

- What is a large language model?
  - Large! (trained on massive dataset)
  - Language! (text! human language)
  - Model! (learned pattern → predict pattern)

It is essentially a collection of numbers/parameters in a specific way, and it is trained on huge amounts of text (books, articles, websites) using deep learning so it can generate fluent, human-like language.

LLMs Basics III

- How large?
  - Old models:
    - 124 million parameters (gpt2)
    - 774 million parameters (gpt2-large)
  - Mid-size models: 7-70 billion (Llama 2)
  - Large LLMs: 70 billion - trillion (gpt 4o)

Week 2 Assignment

- Explain LLMs in different ways: elevator talk, Instagram reels, YouTube shorts, “explain at family Thanksgiving party.”